PC-RadNet (RATF, revised)

LUMBAR SPINE MRI REPORT (draft for radiologist review)
Patient ID: 0021

FINDINGS: At L3-L4, disc protrusion, disc bulge and thecal sac compression are noted, with mild disc degeneration (Pfirrmann grade 2).
At L4-L5, disc protrusion, disc bulge and thecal sac compression are noted, with moderate disc degeneration (Pfirrmann grade 3).
At L5-S1, disc protrusion, disc bulge and thecal sac compression are noted, with advanced disc degeneration (Pfirrmann grade 4).

Estimated disc height (axial regression, coarse): L3-L4 8.7 mm; L4-L5 8.7 mm; L5-S1 5.7 mm.

Disc segmentation confidence (mean mask probability, no reference mask used): L3-L4 0.95; L4-L5 0.95; L5-S1 0.96.

IMPRESSION: L3-L4: Pfirrmann grade 2; disc protrusion, disc bulge, thecal sac compression.
L4-L5: Pfirrmann grade 3; disc protrusion, disc bulge, thecal sac compression.
L5-S1: Pfirrmann grade 4; disc protrusion, disc bulge, thecal sac compression.

MedGemma-4B-IT

Lumbar Spine MRI

1. TECHNIQUE:

MRI of the lumbar spine was performed with [Specify Sequence, e.g., T1-weighted with sagittal and axial reformats]. Sagittal and axial images were obtained.

2. FINDINGS:

L3-L4:

* Disc: Pfirrmann Grade III (inhomogeneous, intermediate gray signal, normal to slightly decreased disc height). Disc height measured 8.9 pixels.
* Findings: Disc protrusion, disc bulge, and thecal sac compression are present.

L4-L5:

* Disc: Pfirrmann Grade IV (inhomogeneous, dark gray to black signal, slightly decreased disc height). Disc height measured 9.0 pixels.
* Findings: Disc protrusion, disc bulge, thecal sac compression, and nerve root compression are present.

L5-S1:

* Disc: Pfirrmann Grade IV (inhomogeneous, dark gray to black signal, slightly decreased disc height). Disc height measured 5.6 pixels.
* Findings: Disc bulge and thecal sac compression are present.

3. IMPRESSION:

There is evidence of degenerative disc disease at L3-L4, L4-L5, and L5-S1. The L4-L5 and L5-S1 discs demonstrate Pfirrmann Grade IV signal changes, indicating significant disc degeneration. These findings are associated with disc protrusion, disc bulge, thecal sac compression, and nerve root compression at L4-L5 and L5-S1. The L3-L4 disc shows less severe degenerative changes.

BioGPT-Large

TECHNIQUE: Lumbar spine MRI was reviewed using the provided structured measurements.

FINDINGS:
- L3-L4: Pfirrmann grade 2; disc height 8.7 pixels; disc protrusion, disc bulge, thecal sac compression.
- L4-L5: Pfirrmann grade 3; disc height 8.7 pixels; disc protrusion, disc bulge, thecal sac compression.
- L5-S1: Pfirrmann grade 4; disc height 5.7 pixels; disc protrusion, disc bulge, thecal sac compression.

IMPRESSION:
- L1 / L2: Pfirr man Grade 0; disc height 8 5 9 pixels; spondylolisthesis < 5mm. - L2 / L3: Pfirmin Gran 0; disc Height 8 5 9px; olisthesis > 3mm. - L3 / L4: Pfirer Grade 0; olisthesis > 4mm. - L4 / L5: Pfirers Grade 0; spondylolisthesis < 1mm.The following is a summary of the findings and recommendations for each measurement in this article: - Pfuir grade: 0 = normal; 1 = mild degeneration; 2 = moderate to severe degenerative disease; 3 = advanced stage with significant instability or deformity. - Discal height: 8 pixel range from 8 to 1 0 0 0 pixels. - Thickness: thickness less than two millimeters indicates early intervertebral disk pathology. - Deformity: any angular change greater than five degrees on sagittal plane imaging. - olisthesis: displacement more than three millimeter at one level or four millimeters at another. - Spondylolisthesis: displacement over ten percent of vertebral body width. - olisthesis above fifty percent of vertebral length. - olisthesis below thirty percent of vertebral lengths. - olisthesis between twenty and forty per cent of vertebral widths. - olisthesis within ninety nine percent.1 + The presence of Modic changes should be noted as it may indicate underlying inflammatory processes.2 + In patients who have undergone previous spinal surgery, there can be increased signal intensity around bone marrow edema that has been present previously but no evidence of recurrent herniation.3 + ▬ ▬ ▬ CONCLUSION ▬ This review provides guidelines for measuring structural abnormalities seen on magnetic resonance images

RADAR

MRI of the lumbar spine is compared to ___. Since ___, the patient has been treated with IV ketorelac. The opacities in the thorax have cleared. The opacities in the cervical spine have also cleared. The opacities in the thorax are probably due to atelectasis. The opacities in the cervical spine are probably due to atelectasis. The opacities in the thorax and cervical spine are improved since ___. The opacities in the thorax are probably due to atelectasis. The opacities in the cervical spine are probably due to atelectasis. The opacities in the thorax and cervical spine are improved since ___. The opacities in the thorax and cervical spine are probably due to atelectasis. The opacities in the thorax and cervical spine are improved since ___. The opacities in the thorax and cervical spine are probably due to atelectasis. The opacities in the thorax and cervical spine are improved since ___. The opacities in the thorax and cervical spine are probably due to atelectasis. The opacities in the thorax and cervical spine are improved since ___. The opacities in the thorax and cervical spine are probably due to atelectasis. The opacities in the thorax and cervical spine are improved since ___. The opacities in the thorax and cervical spine

MAIRA-2

Focal kyphosis is noted at the thoracolumbar junction with compression deformity of the T12 vertebral body. This is better assessed on the concurrently performed T and L-spine radiographs. Multilevel degenerative changes are noted throughout the thoracic and lumbar spine.